

SHAHKAR JAVID

Artificial Intelligence & Software Engineer

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PROFESSIONAL SUMMARY

AI & Software Engineer with an MSc in Artificial Intelligence & Robotics, specialized in LLMs, agentic systems, and applied machine learning in production. Comfortable across the modern AI stack — LLM fine-tuning and RAG, agent orchestration, mechanistic interpretability, computer vision, and time-series forecasting — with a track record of shipping models into real production and closed-loop systems using Python, PyTorch, and cloud/MLOps tooling (AWS, Azure). Currently building on-prem LLM assistants and predictive ML pipelines for GPU/HPC infrastructure.

WORK EXPERIENCE

Artificial Intelligence & Software Engineer

Aug 2025 – Present

GemaTEG SRL | Perugia, Italy

- Designed and built a **local multi-agent AI assistant with LangGraph and LangChain** — pairing a RAG agent for knowledge and documentation queries with a metrics agent equipped with dedicated tools to query, search, plot, and analyze live monitoring data. Runs fully on-prem on the GPUs behind a chat interface, letting engineers interrogate the system in plain language.
- Developed **time-series forecasting models** that predict system trends and drive anticipatory, forecast-driven control, **reducing response time by 25%**.
- Designed and implemented **Linux server-monitoring pipelines with Prometheus & Grafana** to collect, store, and analyze real-time metrics (GPU/CPU utilization, memory, temperature, workload characteristics) at scale.
- Collaborated with hardware teams to build **predictive-maintenance pipelines**, applying ML to anticipate device behavior and optimize response.
- Developed AI-driven **adaptive control combining ML models with classical PID control** for closed-loop optimization, cutting thermal overshoot by 20% and improving energy efficiency.
- Built **data-acquisition and extraction frameworks** to process large-scale telemetry from HPC/GPU systems, feeding real-time analytics and decision pipelines.
- Supported R&D on deploying LLMs in adaptive, closed-loop HPC environments, including performance and power profiling of LLM workloads on GPUs.

Artificial Intelligence Intern

Dec 2024 – May 2025

Radoff | Remote, Italy

- Delivered **end-to-end ML pipelines** for time-series forecasting and anomaly detection — owning data ingestion, preprocessing, training, validation, and deployment.
- Developed and deployed ML solutions on **AWS (EC2, S3, Lambda, SageMaker)** for scalable experimentation and production deployment.
- Applied Agile and Dev/MLOps best practices, including **CI/CD with Git, Docker, and Jenkins**, for robust version control and reproducibility.
- Built automated **ETL pipelines** for ingestion, cleaning, and transformation with Python (Pandas, NumPy) and Apache Airflow.
- Performed advanced feature engineering — time-windowing, lag-feature generation, and outlier treatment — to improve accuracy and generalization.

Artificial Intelligence Developer Intern

Feb 2025 – Mar 2025

BSA Underwriting Agency | Rome, Italy

- Reduced manual document-processing effort by building **LLM-based extraction workflows** (LLaMA, BERT, GPT, Hugging Face Transformers, via REST APIs) that parse policy and claim data from live emails and directories.
- Built automated pipelines to retrieve and extract tabular and textual data from PDF and image documents using **OCR (Tesseract, Google Vision)** and NLP-based parsing.
- Cleaned, structured, and normalized unstructured datasets into business-ready formats, auto-uploading to a centralized portal with high data integrity.

Freelance AI / ML Engineer

Sep 2022 – Aug 2024

Fiverr / Upwork | Remote

- Delivered tailored ML models across **computer vision, anomaly detection, and NLP** for diverse international clients.
- Developed object-detection and classification models for satellite imagery using **YOLO and Faster R-CNN** for high-accuracy geospatial analysis.

- Built enterprise conversational AI, chatbots, and generative models (transformers, seq2seq, GPT) using **Azure OpenAI Service** and REST APIs, integrated with backend document-processing systems.

EDUCATION

MSc in Artificial Intelligence and Robotics Jul 2022 – Apr 2025

Università degli Studi di Roma “La Sapienza” | Rome, Italy

- Specialized in Machine Learning, Computer Vision, NLP, Deep Learning, Retrieval-Augmented Generation (RAG), and Advanced Robotics.
- Projects spanning LLM fine-tuning, generative models, Vision Transformers, object detection/localization, MLOps, and cloud deployment (PyTorch, TensorFlow, PyTorch3D, LangChain, ROS, PDDL).

Bachelor of Electrical Engineering (Electronics) Sep 2014 – Jun 2018

Air University | Islamabad, Pakistan

Foundations in C++/OOP, microcontrollers & microprocessors, control systems, FPGA/VLSI, and digital & analog electronics.

SELECTED PROJECTS

Local RAG Agent — Multi-Tool + Memory [GitHub](#)

Fully local, tool-calling RAG agent built with LangGraph and Ollama: answers questions over PDFs via a Chroma vector store, delegates exact arithmetic to dedicated tools, and persists conversation across restarts with sliding-window memory that keeps the prompt a fixed size. Tools are selected purely by matching the query against tool docstrings — no routing logic, no cloud, no API keys.

Refusal-Direction Interpretability for LLM Safety [GitHub](#)

Locates, measures, and reinforces the internal “refusal” direction of open-weight LLMs (Qwen2.5-1.5B-Instruct) via difference-in-means over residual-stream activations; stress-tests robustness with directional ablation and hardens it through activation addition. Reproduces and extends Arditi et al. (2024) toward defensive AI-safety research.

Multi-Agent Monitoring Assistant (LangGraph + LangChain) [GitHub](#)

Local, multi-agent assistant over live monitoring data, powered by an on-prem Ollama LLM. A metrics agent exposes tools to query, search, plot, and analyze time-series, compressing each series into computed facts (latest, mean, drift) to keep prompts small and grounded in real numbers. Served through a chat interface as both a Grafana-injected widget and a standalone UI.

Multimodal NLP — Visual Word Sense Disambiguation [GitHub](#)

Multimodal word-sense disambiguation using OpenAI’s CLIP to align image–text pairs, outperforming text-only WSD through cross-modal embedding similarity.

Anomaly Detection via Adversarial Training (GANs & Autoencoders) [GitHub](#)

Unsupervised image anomaly detection combining GANs and autoencoders, flagging outliers via reconstruction error and loss heatmaps.

Shape Deformation with 2D Supervision (PyTorch3D) [GitHub](#)

3D mesh-deformation pipeline that optimizes shapes from 2D image guidance via differentiable rendering and custom losses (chamfer, silhouette, edge length).

ROS A* Path Planner (C++) [GitHub](#)

Robust A* path planner in ROS/C++ computing shortest paths over 2D occupancy grids with obstacle avoidance, visualized with RViz markers.

TECHNICAL SKILLS

Languages: Python (NumPy, Pandas, scikit-learn, PyTorch, TensorFlow, Keras, PyTorch3D), C++, SQL, Bash, PDDL, HTML/CSS

LLMs & Generative AI: LLM fine-tuning (BERT, GPT, LLaMA, Qwen), RAG, agentic systems (LangGraph, LangChain), local LLM deployment (Ollama), prompt engineering, mechanistic interpretability

Machine Learning & DL: Supervised & unsupervised learning, CNNs, RNNs, Transformers, Vision Transformers, GANs, autoencoders, object detection (YOLO, Faster R-CNN), anomaly detection, time-series forecasting, NLP (NER, NLI, QA)

Cloud & MLOps: AWS (EC2, S3, Lambda, SageMaker), Azure OpenAI Service & Bot Framework, Docker, Git, Jenkins, CI/CD, FastAPI, REST APIs, cloud-native applications

Tools & Platforms: Hugging Face Transformers, LangChain/LangGraph, Chroma, OpenCV, PyTorch Lightning, Prometheus, Grafana, Linux, ROS/RViz, OCR (Tesseract, Google Vision)

LANGUAGES & CERTIFICATIONS

Languages: English (Native) • Italian (Professional)

Certifications: Microsoft AZ-900 – Azure Fundamentals (In Progress) • Microsoft AI-900 – Azure AI Fundamentals (Planned 2026)